



Artificial Intelligence in Education: An Empirical Analysis of Its Influence on Students' Engagement

SRIDEVI M1

1 Assistant Professor, Department of Commerce and Management, Vijaya College, Jayanagar, Bengaluru

***Corresponding Author: SRIDEVI M**

Received: 03/07/2026 | Revised: 09/07/2026 | Accepted: 15/07/2026 | Published: 17/07/2026

Abstract

The integration of artificial intelligence (AI) in education has transformed traditional learning environments by enhancing accessibility, personalization, and efficiency. This study aims to empirically analyze the influence of AI on students' engagement and learning outcomes in Bengaluru. Using primary data collected from students through a structured questionnaire, the study evaluates how AI-based tools impact academic interaction, participation, and performance. The findings indicate that AI significantly enhances student engagement and improves learning outcomes, although certain challenges such as overdependence and reduced critical thinking were observed. The study contributes to understanding the role of AI in shaping modern educational practices.

Keywords: Artificial Intelligence, Student Engagement, Learning Outcomes, Digital Learning, Education Technology



1. Introduction & Objectives of the Study

AI-powered tools such as chatbots, personalized learning platforms, and recommendation systems have redefined the teaching-learning process in higher education hubs like Bengaluru. This study examines how AI affects students' engagement in learning, academic performance, skill development, and potential dependency on AI tools.



2. Research Methodology & Demographic Profile

A quantitative, descriptive, and analytical design was employed using primary data collected from 110 college students in Bengaluru (105 usable responses: 34.30% Male, 65.70% Female; 64.70% aged 18-19; 71.40% Commerce/Management, 19.00% Science, 9.60% Others) via a 5-point Likert scale questionnaire.



2. Research Methodology & Demographic Profile (Continued)

A quantitative, descriptive, and analytical design was employed using primary data collected from 110 college students in Bengaluru (105 usable responses: 34.30% Male, 65.70% Female; 64.70% aged 18-19; 71.40% Commerce/Management, 19.00% Science, 9.60% Others) via a 5-point Likert scale questionnaire.



3. Likert-Scale & Correlation Analysis

Highest agreement was recorded for "When I am more engaged using AI, my learning outcomes improve" (Mean = 3.95, SD = 0.73), "I sometimes rely on AI instead of thinking independently" (Mean = 3.93, SD = 0.76), and "AI helps me understand academic concepts more effectively" (Mean = 3.92, SD = 0.75). Correlation analysis showed positive relationships for AI Usage <-> Engagement ($r = 0.56$), Engagement <-> Learning Outcomes ($r = 0.52$), AI Usage <-> Academic Performance ($r = 0.48$), and AI Usage <-> Dependency ($r = 0.60$).



3. Likert-Scale & Correlation Analysis (Continued)

Highest agreement was recorded for "When I am more engaged using AI, my learning outcomes improve" (Mean = 3.95, SD = 0.73), "I sometimes rely on AI instead of thinking independently" (Mean = 3.93, SD = 0.76), and "AI helps me understand academic concepts more effectively" (Mean = 3.92, SD = 0.75). Correlation analysis showed positive relationships for AI Usage <-> Engagement ($r = 0.56$), Engagement <-> Learning Outcomes ($r = 0.52$), AI Usage <-> Academic Performance ($r = 0.48$), and AI Usage <-> Dependency ($r = 0.60$).



4. Conclusion

Artificial Intelligence has a significant positive influence on students' engagement and learning outcomes in educational institutions in Bengaluru, while simultaneously presenting challenges related to dependency and reduced independent thinking. Educational institutions should encourage responsible, balanced integration of AI as a supplementary learning resource.



4. Conclusion (Continued)

Artificial Intelligence has a significant positive influence on students' engagement and learning outcomes in educational institutions in Bengaluru, while simultaneously presenting challenges related to dependency and reduced independent thinking. Educational institutions should encourage responsible, balanced integration of AI as a supplementary learning resource.



SRCAA Global Review of Contemporary Research (SGRCR)

Shakti Research Centre and Academia (SRCAA) Bengaluru - 560076, Karnataka, India

E-mail: srcaacontact@gmail.com / admin@srcaa.co.in

Website: www.srcaa.co.in



References

1. World Economic Forum. (2022). The future of jobs report 2022. World Economic Forum.
2. Manyika, J., et al. (2023). Defining the skills citizens will need in the future world of work. McKinsey Global Institute.
3. Singh, A., & Dey, S. (2022). Skill development and employability in India: A study under NEP 2020. *Journal of Education and Practice*, 13(4), 45-52.
4. Gratton, L. (2023). The shift towards continuous learning in organizations. Learning Press.
5. Chakraborty, R., & Misra, P. (2024). Organizational learning culture and innovation performance. *International Journal of Human Resource Studies*, 14(2), 88-102.
6. IBM. (2021). Artificial intelligence in education: Enhancing learning experiences. <https://www.ibm.com>
7. UNESCO. (2021). AI and education: Guidance for policy-makers. UNESCO Publishing.
8. Google. (2023). AI tools in education and their impact on learning. <https://www.google.com>
9. Statista. (2023). Usage of AI tools among students worldwide. <https://www.statista.com>
10. OpenAI. (2023). ChatGPT and AI in education: Opportunities and challenges. <https://www.openai.com>